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CAALM-P: Closed-Loop Anti-Oscillatory Adaptive Non-Invasive Pain Suppression via Reinforcement Learning

PROBLEM STATEMENT

- Chronic pain remains difficult to manage reliably because pharmacological therapies often produce tolerance, systemic side effects, or long-term dependence.
- Most available stimulation systems still operate in open loop, meaning they apply fixed parameters without adapting to the patient's changing signal state.
- This creates an engineering gap: there is no practical non-invasive framework that senses pain-related oscillatory activity and updates stimulation in real time.
- CAALM-P addresses this by formulating pain suppression as a closed-loop signal-processing and control problem rather than a static therapy setting.

PROPOSED SOLUTION

- CAALM-P models a full closed-loop pipeline: biological signal generation, acquisition, preprocessing, feature extraction, classification, adaptive control, stimulation, plant-coupled suppression, and validation.
- The signal passes through an electrode model, amplification and digitization chain, then preprocessing before extracting dominant frequency, envelope, and phase biomarkers.
- A pain-state classifier maps these biomarkers to discrete pain states, and the controller selects anti-phase stimulation parameters adaptively.
- The stimulation waveform generator suppresses the oscillatory component of the next signal block while preserving the residual component through the plant model.
- Post-stimulation metrics are fed back into the loop, enabling controller refinement and repeatable validation.
- The current implementation is single-channel; a lattice-based multi-electrode extension is planned for future spatial modulation.

SYSTEM BLOCK DIAGRAM

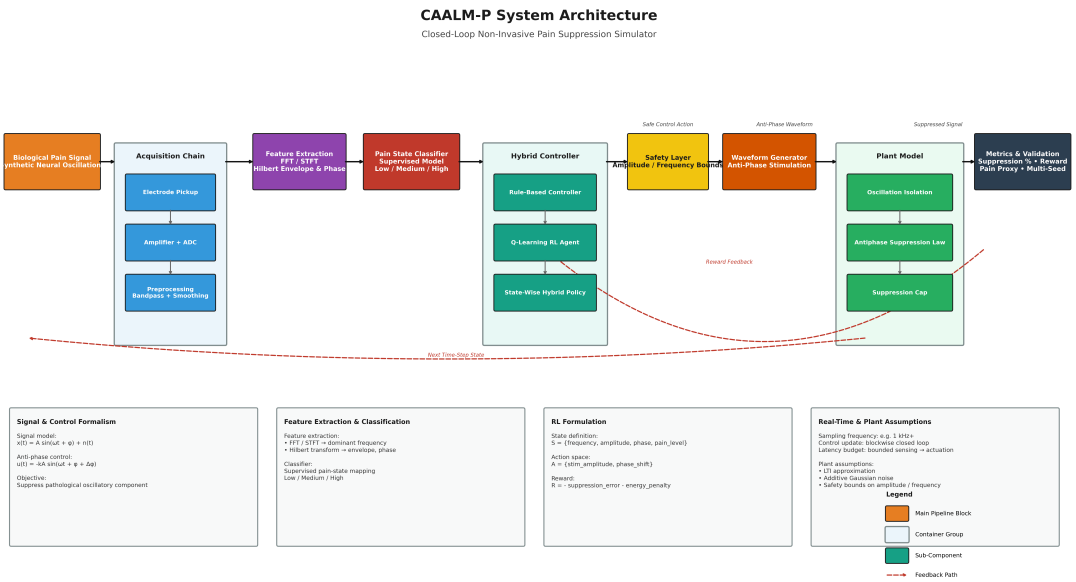

Figure 1: CAALM-P closed-loop signal flow architecture.

SIMULATION RESULTS

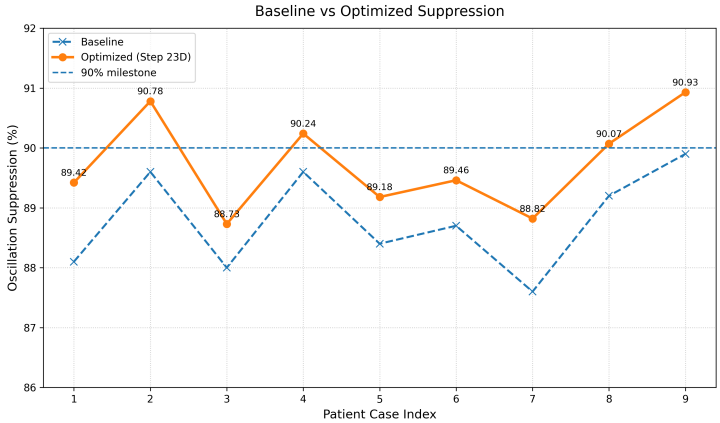

Figure 2: Case-wise benchmark suppression performance under the strongest validated controller.

- The strongest validated benchmark configuration achieved a mean oscillation suppression of approximately 89.74%, with a best single-case suppression of about 90.93%.
- A total of 4 out of 9 benchmark cases crossed the 90% suppression threshold, while all benchmark cases remained above 85%.
- The main evaluation metrics were suppression %, pain-proxy reduction, and controller reward, since together they capture attenuation, therapeutic relevance, and control efficiency.
- During the cap sweep, increasing the suppression cap from 0.95 to 0.99 improved the mean suppression from roughly 88.83% to 89.74%.
- This behavior suggests that the remaining performance limit lies more in plant-side alignment and suppression formulation than in controller search itself.
- An honest limitation is that the present simulator still relies on synthetic pain dynamics and simplified plant assumptions rather than patient-recorded physiological signals.

HARDWARE FEASIBILITY

- A practical prototype would require wearable electrodes, low-noise amplification, ADC digitization, and bounded non-invasive stimulation hardware.
- An STM32-class MCU or similar embedded platform could support filtering, feature extraction, bounded control logic, and real-time stimulation updates.
- The major challenge is maintaining robust phase-aligned stimulation under noisy biosignals while enforcing current-amplitude and frequency safety limits.
- A planned extension is a lattice-based electrode array for spatial modulation, improved pain-field targeting, and robustness across distributed stimulation sites.
- A realistic next step is hardware-in-the-loop testing with recorded biosignals before moving toward fully real-time deployment.

FUTURE WORK

- Extend from single-channel stimulation to a lattice-based multi-electrode architecture for spatial modulation.
- Develop spatial control policies for distributed suppression across electrode grids.
- Validate lattice-based stimulation using hardware-in-the-loop experiments.

WHAT I WANT FROM YOUR LAB

- I want to validate the sensing and feature-extraction pipeline on real physiological recordings available through your lab environment.
- I want to translate the current simulation into a hardware-aware closed-loop prototype using embedded systems and biosignal acquisition infrastructure.
- I especially want to connect the present reward model to hardware-in-the-loop experiments so the project can move from computational validation toward real bioelectronic implementation.

Available for full-time on-campus research from May 2026 to August 2026.